

LARRIS XIE

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EDUCATION

University of Waterloo
Bachelor's of Computer Science

Apr 2028
3.9/4.0 GPA

EXPERIENCE

Conway

Apr 2026 - Present

Software Engineering Intern

- Selected as 1 of 40 **Kleiner Perkins Fellows** from 5000+ candidates
- Created an **durable agent orchestration runtime** for fraud response using the **Claude Agent SDK, Python, and DBOS**, executing workflows over **autoscaling K8s Jobs**.
- Architected a live **SQL rules engine in Python and Snowflake** that compiles stateful SQL into Snowflake Tasks, powering fraud detection across 1.7B+ events at peak volumes of **17M events/day**.

University of Waterloo

Jan 2025 - May 2026

Undergraduate Research Assistant — Security and Privacy in ML

- **Published as a co-first author** [↗](#) on PPHH-VFL, a **privacy preserving inference architecture** for transformer-based **vertical federated learning** on distributed time-series data.
- Developed custom **PyTorch transformer** models with **adversarial GAN privacy training**, achieving a 95% F1 score on fraud detection under a 10,000:1 fraud-to-legitimate transaction ratio.
- Accelerated **private transformer inference** by **up to 44.4×** while retaining 100% of performance by engineering a distributed GPU framework with CrypTen MPC and PyTorch RPC.

Shopify

May 2025 - Aug 2025

Software Engineering Intern

- Orchestrated a **100% uptime data migration** from legacy APIs to new service endpoints while ensuring full **backwards compatibility** through shadow writes, backfilling **over 5B rows**, and staged rollouts.
- Designed and implemented scalable **data models and GraphQL APIs in Ruby on Rails**, powering a fully extensible cash management experience for **2M+ active retailers**.
- Engineered full-stack features for Shopify Point of Sale (POS) in **React Native and Typescript**, resolving critical race conditions in async workflows to reduce failure rate by **40%**.

Meta x MLH

May 2025 - Aug 2025

Production Engineering Fellow

- Deployed and managed **Linux infrastructure** on CentOS, leveraging system calls and kernel-level debugging at the OS layer, and Docker and NGINX at the application layer to improve reliability.
- Configured system monitoring with **Prometheus, cAdvisor, and Grafana**, leveraging low-level resource metrics (CPU, cache misses, memory usage, I/O throughput) to cut bug detection time by **80%**.
- Automated **CI/CD workflows** with GitHub Actions, integrating testing frameworks, security scans, and performance monitoring to streamline deployments and improve system reliability.

PROJECTS

LifeOS (Hack the 6ix Winner) [↗](#) | React, TypeScript, Python, FastAPI, Redis, Docker, PostgreSQL, AWS

- Engineered a **distributed AI video processing pipeline** using Redis Queue that orchestrates semantic analysis with TwelveLabs, embedding generation, and cloud storage via **asynchronous workers**.
- Deployed a **containerized API service with AWS** using Docker, FastAPI, Pydantic models, and JWT middleware; supports modular features like authentication, video ingestion, memory search, and chat.

On The Dot (5000+ users) [↗](#) | React, TypeScript, Python, PostgreSQL, AWS

- Launched a trivia game with persistent game state and anonymous user tracking—**10k plays in 3 months**.
- Architected a Python backend with custom **PostgreSQL** functions to achieve **20ms read/write latency** on **30K queries/day** for real-time statistics tracking and rate control.

ASCII Game Engine [↗](#) C++

- Architected a 2D game engine in C++ from scratch, grounded in **MVC** architecture and **SOLID principles**, and leveraging **advanced design patterns** for modular and efficient resource management.
- Features **OOP** in console-based rendering, event-driven input and audio, collision detection, dynamic world state with multi-layered rendering and animation, with two demo games: Flappy Bird and Space Invaders.